

EGT207 Machine Learning Project

Objective:

To develop ML models to solve a real-world problem with the given dataset, and to apply the relevant techniques to evaluate and optimize the model and efficiency.

- Develop ML models for the given dataset and problem definition.
- Evaluate ML models using appropriate metrics for efficiency and effectiveness.
- Apply model tuning and optimization to improve the performance of ML models.
- Compare the different ML models on their performance with respect to the given dataset and problem definition.

Project Tasks:

1. Dataset: **parkinson_disease_assignment.csv**
2. Develop **Two ML models** for the classification task in a **single Colab notebook**.
 - The first model is **Decision Tree**.
 - The second model is a **self-selected Ensemble Method**. You are expected to select an appropriate Ensemble Method based on the given dataset and task.
 - Add your **adminNo** as the prefix of all ML models' name.
3. Manage the Colab notebook with below sections:
 - Data preparation
 - Build Decision Tree model
 - Summary of Decision Tree model development (<150 words): to describe your key considerations and concerns on design, evaluation, optimization, regularization, etc.
 - Build Ensemble Method model
 - Summary of Ensemble Method model development (<150 words): to describe your key considerations and concerns on model selection, design, evaluation, optimization, regularization, etc.
 - Comparison of the 2 models (Decision Tree and Ensemble Method) (<200 words): to compare and explain their results and performance, pros/cons, suitability for the given task, any other concerns.
 - Keep all running results within the notebook.

Submission:

- Submit a single Colab notebook ([admin_project.ipynb](#)) via Project Submission (Week17) link in Brightspace.
- Deadline: By end of Week 17 (23:59 on 11 Aug 2024).

*This is an individual project. **ZERO** mark will be given if fully copy is found!

Assessment Rubrics

Marks will be allocated based on the task completeness, model development, techniques, model performance and the summary and comparison:

Items	Need Improvement	Satisfactory	Excellent
Data Preparation (15%)	Poor preparation: • data review and visualization • data correction • data splitting • data engineering	Satisfactory preparation: • data review and visualization • data correction • data splitting • data engineering	Excellent preparation: • data review and visualization • data correction • data splitting • data engineering
Decision Tree (20%)	Poor development: • training • evaluation • optimization and regularization • performance and results	Satisfactory development: • training • evaluation • optimization and regularization • performance and results	Excellent development: • training • evaluation • optimization and regularization • performance and results
Summary of Decision Tree (10%)	Insufficient and unclear description and explanation: • evaluation and optimization • overfitting techniques • results and performance	Insufficient and clear description and explanation: • evaluation and optimization • overfitting techniques • results and performance	Sufficient and clear description and explanation: • evaluation and optimization • overfitting techniques • results and performance
Ensemble Method Model (20%)	Poor development: • selection • training • evaluation • optimization • overfitting techniques • performance and results	Satisfactory development: • selection • training • evaluation • optimization • overfitting techniques • performance and results	Excellent development: • selection • training • evaluation • optimization • overfitting techniques • performance and results
Summary of Ensemble Method model (10%)	Insufficient and unclear description and explanation: • model selection • evaluation and optimization • overfitting techniques • results and performance	Insufficient and clear description and explanation: • model selection • evaluation and optimization • overfitting techniques • results and performance	Sufficient and clear description and explanation: • model selection • evaluation and optimization • overfitting techniques • results and performance
Comparison of 2 models (15%)	Comparisons are not clear: • results and performance • pros/cons • reasoning for decisions • comprehensive views and supporting evidence • technical knowledge • understanding of problem	Comparisons are insufficient: • results and performance • pros/cons • reasoning for decisions • comprehensive views and supporting evidence • technical knowledge • understanding of problem	Comparisons are clearly organized and conveyed: • results and performance • pros/cons • reasoning for decisions • comprehensive views and supporting evidence • technical knowledge • understanding of problem
Execution Performance (10%)	Poor execution performance: • errors in notebook • execute correctly • notebook developed in sequences • originality & creativity • new ideas or insights	Satisfactory execution performance: • errors in notebook • execute correctly • notebook developed in sequences • originality & creativity • new ideas or insights	Excellent execution performance: • errors in notebook • execute correctly • notebook developed in sequences • originality & creativity • new ideas or insights